

(a, b, d) same as before (e) no.

$$3-3)(e) k' \lambda m' + k' m' n + k \lambda m' n' + \lambda m n'$$

$$= k' \lambda m' (n + n') + k' m' n (\lambda + \lambda') + k \lambda m' n' + \lambda m n' (k + k')$$

$$= k' \lambda m' n + k' \lambda m' n' + k \lambda m' n + k' m' \lambda' n + k \lambda m' n' + k \lambda m n' + k' \lambda m n'$$

$k\lambda \backslash mn$	00	01	11	10
00		1		
01	1	1		1
11	1			1
10				

$$F = k' m' n + \lambda n' + k' \lambda m'$$

1	1	1	1
	1		

Ch-3

3-1. (b) $F(A, B, C, D) = \sum (7, 13, 14, 15)$

AB \ CD	00	01	11	10
00				
01			1	
11		1	1	1
10				

$$F = BED + ABD + ABE$$

(a) (a, c, d) same as before (b) no.

3-2) (c) $a'b' + be + d'be'$

$$= d'b'(e+e') + abe(a+a') + a'be'$$

$$= a'b'e + a'b'e' + abe + a'be + a'be'$$

ab \ cd	00	01	11	10
0	1	1	1	1
1			1	

$$F = a' + be$$

(a, b, d) same as before (e) no. =

$$3-3] (e) k' l m' + k' m' n + k l m' n' + l m n'$$

$$= k' l m' (n + n') + k' m' n (l + l') + k l m' n' + l m n' (k + k')$$

$$= k' l m' n + k' l m' n' + k l m' n + k' m' l' n + k l m' n' + k l m n' + k' l m n'$$

mn \ kl	00	01	11	10
00		1		
01	1	1		1
11	1			1
10				

$$F = k' m' n + l n' + \cancel{k l m'}$$

$$(d) A'B'e'd' + Ae'd' + B'eD' + A'BeD + Be'd$$

$$= A'B'e'd' + Ae'd'(B+B') + B'eD'(A+A') + A'BeD +$$

$$Be'd(A+A')$$

$$= A'B'e'd' + AB'e'b' + AB'e'd' + AB'eD' + A'B'eD' +$$

$$+ A'BeD + AB'eD + A'Be'd$$

AB \ CD	00	01	11	10
00	1			1
01		1	1	
11	1	1		
10	1			1

$$F = B'eD' + A'BeD + AB'e'$$

(a, b, e) same as before ~~be~~(e, d) no -

$$(3-5) (a) F_1 = \prod (0, 3, 5, 6)$$

$$= (x+y+z) (x+y'+z') (x'+y+z') (x'+y'+z)$$

$$= M_0 M_3 M_5 M_6$$

$$F_2 = \prod (0, 1, 2, 4)$$

$$= (x+y+z) (x+y+z') (x+y'+z) (x'+y+z)$$

$$= M_0 M_1 M_2 M_4$$

$$(b) F_1 = \sum (1, 2, 4, 7)$$

$$F_1 = x'y'z + x'yz' + xy'z' + xyz$$

$x \backslash yz$	00	01	11	10
0		1		1
1	1		1	

$$F_2 = \sum (3, 5, 6, 7)$$

yz	00	01	11	10
x			1	
0			1	
1		1	1	1

$$F = yz + xz + xy$$

(c) From (b) we get

$$F_1 = \prod (0, 3, 5, 6) \text{ POS}$$

yz	00	01	11	10
x				
0	0	0	0	0
1		0		0

$$F = (x' + y + z) (x + y' + z') (x' + y + z')$$

$$(x' + y' + z)$$

$$F_2 = \prod (0, 1, 2, 4) \text{ POS}$$

yz	00	01	11	10
x				
0	0	0		0
1	0			

$$F = \frac{x'z' + x'y' + y'z'}{(x+z)(x+y)(y+z)}$$

3-6 (b) $F(A, B, C, D) = \Pi(0, 1, 2, 3, 9, 10, 11)$

AB \ CD	00	01	11	10
00	0	0	0	0
01	0			
11				
10			0	0

$$F = \cancel{A' C' D' + A' B' + B' C} \\ = (A + C + D)(A + C + D)(A + B)(B + C')$$

(a, c) same as before (b) no. -

(3-7) (a) $x'z' + y'z' + \bar{y}z' + xyz$

$$= x'z'(y + y') + y'z'(x + x') + (x + x')yz' + xyz$$

$$= \overset{1}{x'y'z'} + \overset{2}{x'y'z'} + \overset{3}{xy'z'} + \overset{2}{x'y'z'} + \overset{4}{xyz'} + \overset{2}{x'y'z'} \\ + \overset{5}{xyz}$$

$$= x'y'z' + x'y'z' + xy'z' + x'y'z' + xyz$$

(1) SOP $\rightarrow$ 1

	00	01	11	10
0	1			1
1	1		1	1

$$y = z' + xy$$

pos $\rightarrow 0$

	00	01	11	10
0		0	0	
1		0	1	

$$F = \sum m(2, 3, 4, 5, 6, 7)$$

$$= (x + z')(y + z)$$

$$(b) (A + B' + D)(A' + B + D)(C + D)(C' + D')$$

$$= (A + B' + D + C)(A' + B + D + C')$$

$$= (A + B' + D + C)(A' + B + D + C')$$

$$A + B' + D + C = (A + B' + C + D)(A + B' + C' + D)$$

$$A' + B + D + C' = (A' + B + C + D)(A' + B + C' + D)$$

$$(C + D + A A' + B B') = A + C + D$$

$$(b) (A+B'+D) (A'+B+D) (C+D) (C'+D')$$

$$= (A+B'+D+CC') (A'+B+D+CC') (C+D+AA') (C'+D'+AA')$$

1st term -

$$(A+B'+C+D) (A+B'+C'+D)$$

2nd term -

$$(A'+B+C+D) (A'+B+C'+D)$$

3rd term -

$$(A+C+D) (A'+C+D) = A \cdot$$

$$= (A+C+D+BB') (A'+C+D+BB')$$

$$= (A+B+C+D) (A+B'+C+D) (A'+B+C+D) (A'+B'+C+D)$$

4th term -

$$(A+C'+D') (A'+C'+D')$$

$$= (A+C'+D'+BB') (A'+C'+D'+BB')$$

$$= (A+B+C'+D') (A+B'+C'+D') (A'+B+C'+D') (A'+B'+C'+D')$$

Combining all terms and removing those that appear more than once -

$$= (A+B'+C+D) (A+B'+C'+D) (A'+B+C+D) (A'+B+C'+D) (A+B+C+D) (A'+B'+C+D) (A+B+C'+D) (A'+B+C'+D) (A'+B'+C'+D)$$

(2) POS $\rightarrow$ 0

AB \ eD	00	01	11	10
00	0		0	
01	0		0	0
11	0		0	
10	0		0	0

$$F = (e+d)(e'+d')(A+B'+D)(A'+B+D)$$

(1) SOP $\rightarrow$ 1

AB \ eD	00	01	11	10
00		1		1
01		1		
11		1		1
10		1		

$$F = e'D + A'B'eD' + ABCD'$$

(e, d, e) same as before

3-15 (b) $F = B'c'd' + Bcd' + ABCD'$

$d = B'cd' + ABc'd$

$$F = B'c'd'(A+A') + Bcd'(A+A') + ABCD'$$

$$= AB'c'd' + A'B'c'd' + ABcd' + A'Bcd' + ABCD'$$

$d = B'cd'(A+A') + ABc'd$

$$= AB'cd' + A'B'cd' + ABc'd$$

Combining 1's and X's

$AB \backslash CD$	00	01	11	10
00	1			X
01	1		1	1
11		X		1
10		X		X

For

$AB \backslash CD$	00	01	11	10
00	1			X
01				1
11		X		1
10	1			X

$$F = B'D' + CD$$

(a) same as before -

3-16 (a) $F = AB'D' + A'CD + ABC$

$$= A'B'D'(C+C') + A'CD(B+B') + ABC(D+D')$$

$$= A'B'D'CD + A'B'D'C'D' + A'BCD + A'B'CD + ABCD + ABCD'$$

$$= A'B'D' + A'CD + ABC$$

$$= A'B'D' + A'CD + ABC$$

$$= A'B'D' + A'CD + ABC$$

AB \ CD	CD			
	00	01	10	11
AB	00	1	1	1
	01		1	1
	10			
	11			

AB \ CD	CD			
	00	01	10	11
AB	00	0	1	0
	01	0	0	1
	10	1	0	1
	11	0	1	0

$F = B'D' + A'CD$ Sum of product

$$F = B'D' + A'CD$$

2) POS

$$F = \cancel{cD'} + \cancel{A'B'} + \cancel{B'e}$$

$$= (c + D') (A' + B') (B' + e)$$

(b, e) same as before.

3-4) (c) $A'B'cE' + A'B'e'D' + B'D'E' + B'cD' +$
 $cDE' + BDE'$

$$= A'B'cE'(D + D') + A'B'e'D'(E + E') + B'D'E'(A + A')(e + e')$$

$$+ B'cD'(A + A')(E + E') + cDE'(A + A')(B + B') +$$

$$BDE'(A + A')(e + e')$$

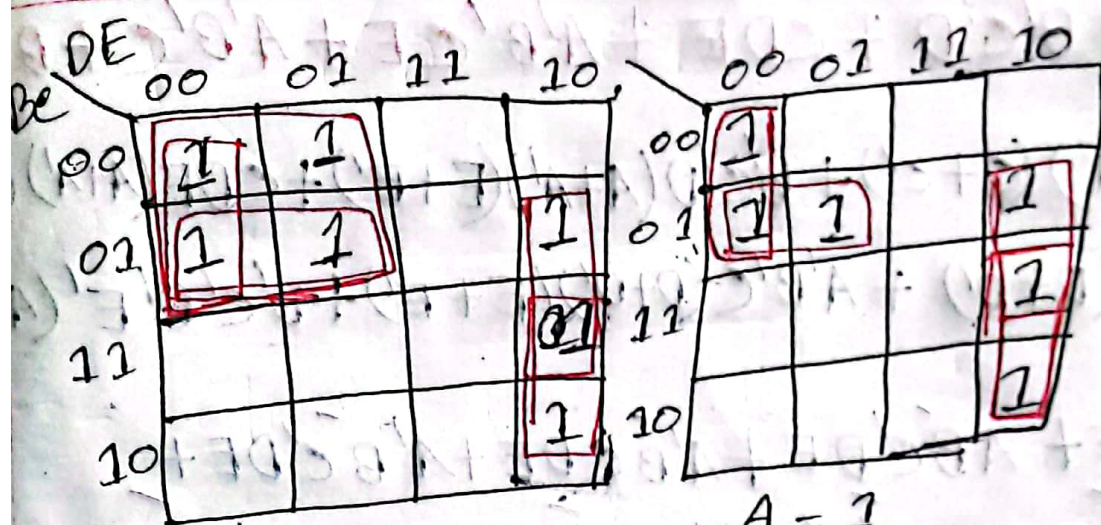
$$= A'B'cDE' + A'B'cD'E' + A'B'e'D'E + A'B'e'D'E' +$$

$$AB'cD'E' + AB'e'D'E' + A'B'cD'E' + A'B'e'D'E' +$$

$$AB'cD'E' + AB'e'D'E' + A'B'cD'E' + A'B'e'D'E' +$$

$$AB'cDE' + AB'eDE' + A'B'cDE' + A'B'eDE' +$$

$$AB'cDE' + AB'eDE' + A'B'cDE' + A'B'eDE'$$



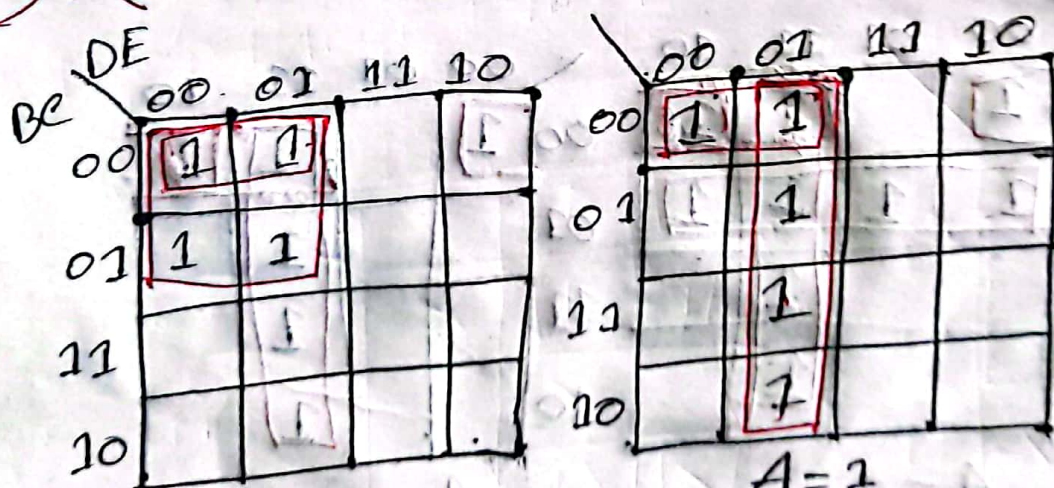
A=1

A=0

$$F = \overline{C}DE' + BDE' + B'C'D' + B'D'E' + B'$$

$$A'B'D'$$

(Q) $F(A, B, C, D, E) = \sum (0, 1, 4, 5, 16, 17, 21, 25, 29)$



A=0

A=1

$$F = B'C'D' + A'B'D' + AD'E$$

$$(b) BDE + B'c'D + cDE + A'B'c'E + A'B'c + B'c'D'E'$$

$$= BDE(A+A')(c+c') + B'c'D(A+A')(E+E') + cDE(A+A')(B+B') + A'B'cE(D+D') + A'B'c(D+D')(E+E') + B'c'D'E'(A+A')$$

$$= ABCDE + ABc'DE + A'BcDE + A'Bc'DE + AB'c'DE + AB'c'DE' + A'B'c'DE + A'B'c'DE' + ABCDE + ABc'DE + A'BcDE + A'Bc'DE + A'B'cDE + A'B'cDE' + A'B'c'DE + A'B'c'DE' + AB'c'DE' + A'B'c'DE'$$

BE/DE

	00	01	11	10
00	1		1	1
01	1	1	1	1
11			1	
10			1	

A=0

	00	01	11	10
00	1		1	1
01			1	
11			1	
10			1	

A=1

$$F = DE + B'c'E' + A'B'c$$

A₂